

Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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Abstract

Resampling priors can steer *where* a differentiable triangle-soup reconstruction spends its triangle budget, but our controlled study of that channel (reported in Appendix A) found their topological value is carried largely by spatial *width*, and that no prior shape repairs one-dimensional features: concentrated priors manufacture phantom handles, spread priors never beat a random control. We move topology from the allocation channel into the *objective*: a differentiable persistence loss $L = \text{photometric} + \lambda(t) L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface’s persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward pass: the matched birth/death simplices are re-expressed as closed-form circumradii of live sample positions — exact for the Gabriel simplices that constitute all significant pairs in practice — and a recruitment term restores the gradient that optimal matching provably lacks when a target feature is missing entirely. A single ratio knob ρ sets the loss scale by gradient-norm calibration; a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve 4.0–7.9 $\times$ and — the channel’s distinguishing result — torus loops improve 2.3 $\times$ with zero phantom handles across seeds, while the control collapses a loop. The two channels compose: loss plus the best resampling prior reaches 7.6 $\times$ below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — a third channel agreeing that this class needs no topological information. All verdicts replicate, with no per-shape tuning, on four external genus-known meshes — one of them a raw artist mesh ingested and *certified* rather than constructed — at 2.1–10.4 $\times$ and equal-or-better Chamfer, and the loss degrades gracefully — never below baseline — under sensor noise up to one sample spacing. Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1 Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [1], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff distances while the bottleneck distance between persistence diagrams separates them by an order of magnitude (Appendix A). We first attacked this through the *allocation* channel — biasing where the resampler spends triangles — in a controlled study

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reported with its key numbers in Appendix A. The verdict was mixed: a topological prior helps enclosed voids, but most of the gain is reproduced by any equally-wide non-topological prior, and no prior shape repairs one-dimensional features: concentrating manufactures phantom handles, and spreading never beats a random control on loops.

This paper moves topology into the *objective*. We add a differentiable persistence term to the training loss, so gradients act on the birth/death values of the reconstruction’s topological features directly, rather than on where triangles respawn. Concretely:

- **A matched persistence loss with a pair-frozen backward (§3):** the forward pass is ordinary exact persistence (GUDHI alpha complex); the backward pass re-expresses each paired birth/death simplex as the closed-form circumradius of live surface samples. On our shapes, 100% of significant pairs are Gabriel-exact, so the frozen gradient is the true gradient almost everywhere.
- **A recruitment term for missing features:** optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode. Recruitment attaches the nearest live bar instead, restoring a descent direction; a controlled toy probes its known location-blindness, and an in-loop ablation shows the term is load-bearing — removing it collapses the loop-class win to baseline (§5).
- **One calibrated knob:** λ is set by a gradient-norm ratio ρ against the photometric loss. The response is flat over a $10\times$ range of ρ , and a curriculum schedule adds nothing once calibrated — an ablation, not an assumption.
- **Channel-controlled evaluation:** every claim is tested against a norm-matched *non-topological* control acting through the identical channel (a repulsion on the same surface samples), inheriting the control discipline that made the allocation-channel study conclusive.

The loss is topology-specific for enclosed voids ($4.0\text{--}7.9\times$ error reduction at Chamfer parity) and, unlike every prior shape we tested, for loops ($2.3\times$, zero phantom handles across seeds). The two channels compose (best-of-both $7.6\times$), and component structure (H0) is once again matched by the generic control — three channels now agree that this class carries no exploitable topological signal.

2 Related work

Differentiable rasterization and soup pipelines. OpenDR [2], neural mesh rendering [3], SoftRas [4] and nvdiffrast [5] established gradient flow from pixels to geometry; Gaussian splatting [6] and triangle splatting [7] trade connectivity for speed. The pipeline we build on is DiffSoup [1], whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (Appendix A) steered with importance priors; here the resampler stays untouched and the objective changes instead. Vertex-domain optimizers such as large-steps preconditioning [8] are complementary — they change *how* vertices move, we change *why*.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper’s problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal *by construction*, the discriminating-dimension bottleneck separates correct from broken topology by $35\text{--}40\times$ (Table 3). (2) For voids, a spread topological prior cuts the bottleneck $33\text{--}53\%$ below baseline — but a width-matched *random* control recovers most of that gain, leaving

only a small, shape-dependent topological residual: the prior’s value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the *non-topological* wide control (.0267 vs the topological spread prior’s .0316), and the concentrated topological field manufactures phantom handles (4.4 significant loops against the true 2). The composition condition C5 reuses that study’s strongest field (B4, defined in Appendix A), but the mechanism is not specific to its construction — any width-matched allocation prior can fill the slot.

Persistence in learning objectives. Persistent homology [9] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [11], for topology-preserving image segmentation [12, 13], point-cloud and shape optimization [14, 15], and diagram-metric optimization with convergence analysis [16]; barcode-valued maps have since been given a general differential framework [17], and the per-step cost of persistence-guided descent attacked directly [18]. Our backward pass is in this family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex’s birth/death as a closed-form circumradius of live surface samples — with three pipeline-specific ingredients. (i) *Alpha-complex pairing over $\alpha \times \text{area}$ surface samples* of a live triangle soup, rather than image grids or raw clouds, couples the diagram to the geometry the renderer is actually training. (ii) A *Gabriel-exactness gate*: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — the gate is orthogonal to the standard pair-frozen device and certifies when the frozen gradient *is* the true gradient, rather than assuming it. (iii) *Recruitment for missing features*: optimal partial matching has zero gradient exactly when a target feature is entirely absent — the worst failure mode — so we deliberately trade metric fidelity for a restored descent direction (§3), and quantify both its known pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [19–21], and budgeted remeshing has a long lineage [22–26]. Our contribution is orthogonal: the budget mechanics stay DiffSoup’s; only the training signal becomes topology-aware.

3 Method: a matched persistence loss for triangle soups

3.1 Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [1], which we adopt *unchanged*: every ablation below isolates the topology channel, never photometric tuning. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed *target bundle* $\mathcal{T}$: the significant finite bars of the ground-truth shape’s diagram, computed once at the same sampling density M the loss will see (alpha birth/death values shift with point spacing, so density matching is a correctness requirement, not a convenience) and normalized by the target’s bounding-box diagonal, which is locked for all live evaluations.

3.2 Live diagram and pair-frozen backward

At each refresh we draw $M=2048$ surface samples $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha) \times \text{area}$ -weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step,

so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X yields persistence pairs (birth simplex, death simplex) per dimension. For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar’s coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$R_{\text{edge}} = \frac{1}{2} \|p_1 - p_0\|, \quad R_{\text{tri}} = \frac{\|u\| \|v\| \|u-v\|}{2 \|u \times v\|}, \quad R_{\text{tet}} = \|y\| \text{ with } Ay = \frac{1}{2} \text{diag}(AA^\top), \quad (2)$$

where $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}). A pair failing this Gabriel test is *detached*: its term keeps contributing its (correct) value to the loss but propagates no gradient through that simplex — an exactness failure can therefore never inject a wrong gradient or destabilize training; the worst case is a missing tug for one refresh interval, re-tested at the next re-pairing. On all six benchmark shapes 100% of significant pairs are Gabriel-exact on clean clouds, and the sensor-noise stress test (§4) never triggered the fallback — zero failures across 2,400 noisy refreshes (§5). Alternatives compatible with the same gate — dropping failing pairs outright, subgradient surrogates, or locally raising M near degeneracies — exist; detachment is the most conservative, and the per-refresh failure count is logged so practitioners can judge when a stronger fallback is warranted. The pairing is treated as locally constant — the standard pair-frozen device [11, 16] — and refreshed every $K=10$ steps and, necessarily, immediately after every resampling event, which replaces the vertex and face tensors wholesale.

3.3 Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] + \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2}\right)^2 + L_{\text{recruit}}. \quad (3)$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot: when a target feature is entirely missing, the cheapest transport diagonalizes both sides and the gradient is exactly zero — precisely in the most important failure case. Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, *sub-threshold* bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth–death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} [(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2], \quad k(j) = \arg \min_{k \in \mathcal{A}_j} [(b_k - b_j^*)^2 + (d_k - d_j^*)^2], \quad (4)$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term. Both coordinates of the recruited bar back-propagate through the same circumradius machinery as matched bars. This deliberately departs from a metric-faithful W_2 ; §5 probes its known pathology (location-blindness) directly, and the C6 ablation shows the term is load-bearing in training.

3.4 Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless; $\rho=0.1$ throughout. The tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an uncalibrated-schedule ablation (constant λ from step one) matches the ramped version — calibration, not scheduling, is what makes the knob safe. The window itself carries no measurable sensitivity either: alternative ramps (10–40% and 30–60% of training; sphere, three seeds) land at $.0154 \pm .0017$ and $.0179 \pm .0019$ against the default window’s $.0156 \pm .0013$ — within seed noise. The 20:50 default is a conservative choice (let photometric pressure establish coarse geometry before topology guidance activates), not a tuned hyperparameter.

3.5 Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: $+$ a short-range repulsion on the *same* frozen samples, calibrated with the *same* ρ — a norm-matched non-topological control through the identical channel (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 combined with the best prior of our allocation-channel study (the spread topological resampling field B4; defined in Appendix A). C6: C1 with the recruitment term removed (pure matching $+$ diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair? The hook adds one loss line to the training loop; with the flag off, no new code executes, and a $\rho=0$ run is divergence-equivalent to baseline re-runs under CUDA nondeterminism.

4 Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (the loader’s standard every-8th test hold-out leaves 21–63 training views): sphere and cube (void class H2, $\beta=(1, 0, 1)$, budget $N=1200$), torus (loop class H1, $(1, 2, 1)$, $N=700$), two spheres (component class H0, $(2, 0, 2)$, $N=700$), and genus-2 double torus as a stretch case $((1, 4, 1)$, $N=2000$). Budgets are the allocation study’s head-room points (Appendix A) — tight enough that the baseline’s topology is imperfect, so there is something to win. Four external, genus-known meshes probe generality beyond the analytic family (§5.3). Three are watertight benchmarks: *spot* (genus 0) and *bob* (genus 1) from the Crane model repository (CC0), and the *fandisk* CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px), class budgets carried over unchanged ($N=1200$ voids, $N=700$ loops), and a seed-0 C0 pilot confirming baseline imperfection (tail bottleneck $.043$ – $.054$, i.e. inside the analytic shapes’ band). The fourth, the *tom-yum pot* (Fig. 6), probes the step every real deployment needs: starting from a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup — 6,318 vertices in 9 overlapping open shells, mixed quad/12-gon faces, 50 boundary edges after exact welding, 232 non-manifold junction edges — with no trustworthy ground-truth topology (the same defect that excluded ShapeNet). An ingest-and-certify pass makes it usable: exact-duplicate weld, offset-solidification into a thin closed shell (exact unsigned distance field $+$ marching cubes at iso ε), dropping 804 enclosed pocket shells, then *certification by measurement* — a closed, edge-manifold, consistently-oriented edge certificate, exact simplicial homology, a watertightness cross-check, and bit-identical rebuilds — yielding one body of genus 3, $\beta=(1, 6, 1)$ exact (190,990 vertices), rendered and budgeted as a void-class shape ($N=1200$, 48 views, 200 px). Where the benchmark trio’s topology is known by construction, the pot’s is known by certificate — the pipeline real scans will need. The pot also exercises the measurement floor of §6 prospectively:

every H1 loop of the certified shell (handle bores, flue) sits below the $6r_{\text{med}}$ floor at every working M , while the narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber — an H2 void localized on the pot axis. The void first clears the floor at $M=8192$ (margin $1.20\text{--}1.23\times$ across five sampling seeds), so its target bundle is built there by the pre-registered floor rule — the study’s only protocol deviation — and the loss samples the live surface at the same density (the density-matched contract; a no-op at the default $M=2048$ everywhere else). The bundle carries exactly one significant bar (H2), stable to re-sampling ($\sim 10^{-5}$).

Protocol. 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5, plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. A sensor-noise stress test (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss *plan* sees with i.i.d. Gaussian noise at every refresh, $\sigma = 0.5\%$ and 1% of the target diagonal (≈ 0.5 and $1.0\times$ the median sample spacing at $M=2048$); the pairing, Gabriel gates, and recruitment decide on the noisy observation while the pulls apply to the true live positions. at $M=2048$, and a loss must not act on features it cannot reliably measure at its sampling density. For the same reason the double torus targets its two measurable loops (of four) — matching what the reconstruction itself can express at any feasible budget. The generality wave runs C0/C1/C2 with three seeds; bob, unlike the thin torus, has both its loop pair *and* its void significant at $M=2048$ (fat tube), so it runs the full unrestricted bundle; the tom-yum pot runs at the floor-rule density above.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape’s discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. *Verdict rule:* C1 must beat C0 **and** every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. *Chamfer parity*, formally: the passing arm’s seed-mean Chamfer must satisfy $\text{Ch}(C1) \leq 1.15 \text{Ch}(C0)$; the 15% headroom absorbs seed variance and CUDA nondeterminism, and in practice it never did any work — every passing arm’s Chamfer was at or below the baseline’s. diverge at the first resample), so no bit-level claims are made anywhere.

5 Results

5.1 Gate: the loss alone repairs topology

Before integration, the loss (no photometric term, Adam directly on point positions) must repair defective clouds. It does, on all four probes (Fig. 1): a punctured sphere’s void birth is pulled to target ($8.8\times$ bottleneck reduction; the hole closes), a spurious handle is pushed to the diagonal and dies (H1 bar eliminated; β restored), mis-spaced components are pulled to the target separation ($1229\times$), and — the recruitment probe — a bridged merge with *no* significant live H0 bar is severed ($450\times$, $\beta_0 1\rightarrow 2$). The predicted location-blind pathology did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar’s death monotonically toward target — the runaway concentrates on the only profitable direction.

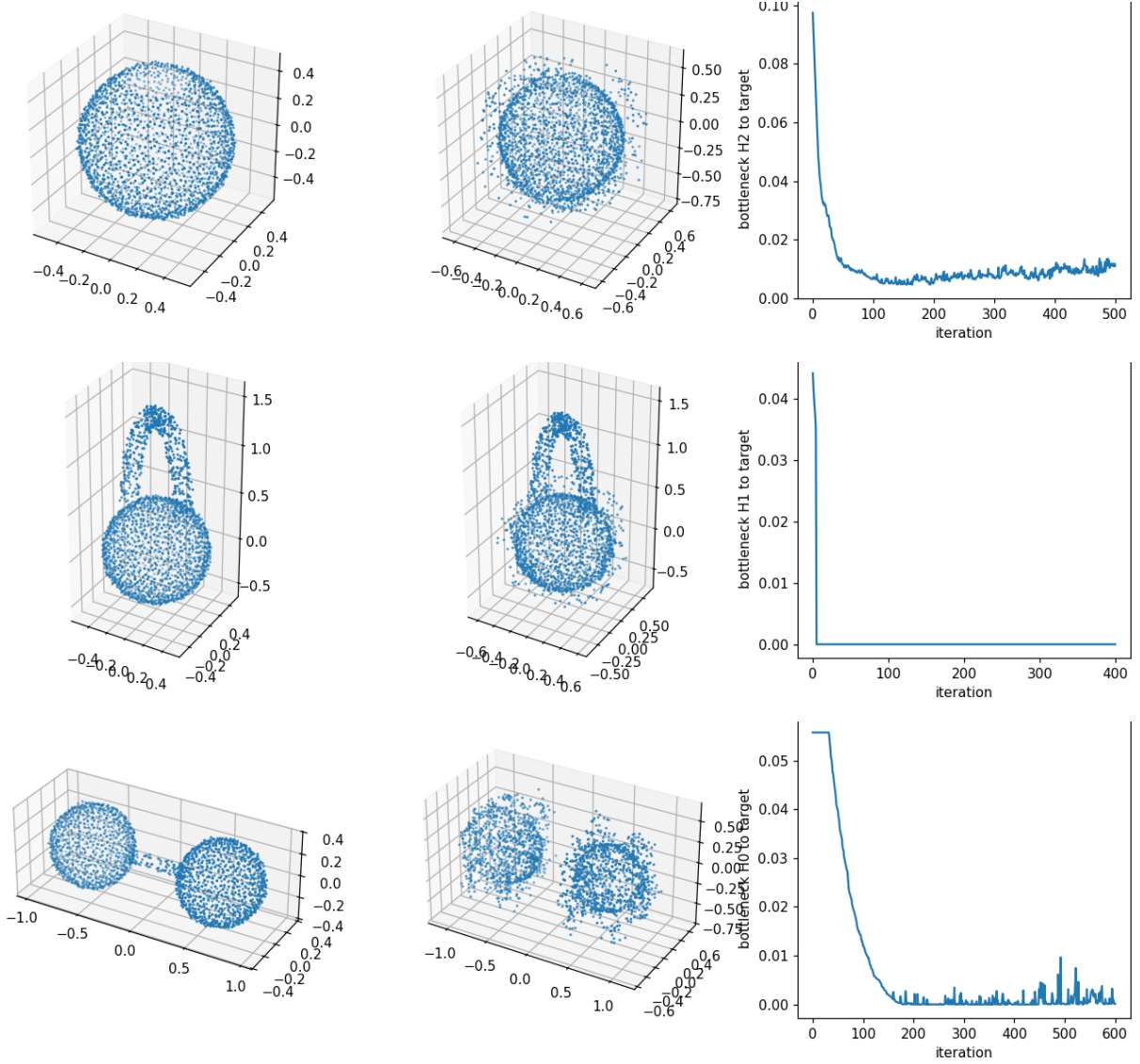


Figure 1: The stage-3a gate: the loss alone (Adam on raw point positions, no photometric term) repairs defective clouds. Each row shows the initial cloud, the final cloud, and the bottleneck-to-target trajectory in the discriminating dimension — one row per mechanism of the matched loss (§3). *Top* (T1): punctured sphere; the matched birth term pulls the void closed ($8.8\times$ reduction). *Middle* (T2): spurious handle; the diagonal term kills the extra H1 bar outright. *Bottom* (T4): bridged merge with no significant live H0 bar; recruitment severs the bridge ($450\times$, β_0 1 $\rightarrow$ 2) — the case where plain optimal matching provably has zero gradient. The mis-spaced components probe (T3, 1229 $\times$) is omitted for space.

Table 1: Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict requires C1 < C0 and C1 < every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623 $\pm$.0063	.0156$\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082$\pm$.0007 (7.6 $\times$)
cube (H2)	.0582 $\pm$.0046	.0074$\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—
torus (H1)	.0424 $\pm$.0031	.0180$\pm$.0016 (2.3 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0015	.0133$\pm$.0004 (3.2 $\times$)
two_spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.5 $\times$)	.0008 (ties C1)	—	—
double_torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—

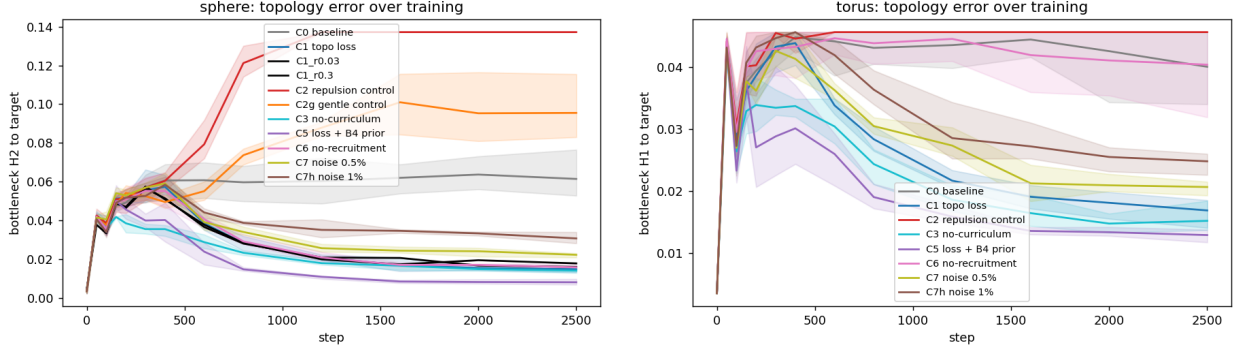


Figure 2: Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min-max). *Left*, sphere (H2): every loss arm — C1, C3, C5, C6, both ρ variants, and the noise-stress pair C7/C7h — converges to a fraction of baseline, while the norm-matched repulsion control C2 is *destructive* and even its gentle variant C2g ends worse than baseline. *Right*, torus (H1), the class no resampling prior won: same ordering, with C5 (loss + spread prior) best; the noise arms descend shallower with more noise but never approach baseline — except the no-recruitment ablation C6, which rides it (see *Ablations*).

5.2 The C-matrix

Table 1 reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; Fig. 2 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere 4.0 $\times$ (16.2 σ Welch vs C0), cube 7.9 $\times$ (18.5 σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is not merely weaker — it is *destructive*: it pins the void’s death at its repulsion equilibrium (bottleneck 2.2 $\times$ worse than baseline, identical across seeds) and on the cube erases the void outright. Nor is this an artifact of control strength: a gentle variant at half the push radius still lands 1.6 $\times$ *worse* than baseline (.0972 vs .0623) — the sphere ordering is C1 < C0 < C2g < C2, so no tested repulsion strength reproduces any of the gain (Fig. 3); Appendix B visualizes the mechanism as diagram trajectories — the control *pins and drags* the void’s death while phantom bars accumulate.

Loops (H1): the channel’s distinguishing result. In the allocation-channel study (Appendix A) no prior shape won this class — concentration manufactured phantom handles; spreading never beat a random control. The loss cuts torus H1 error 2.3 $\times$ with *zero phantom handles in all*

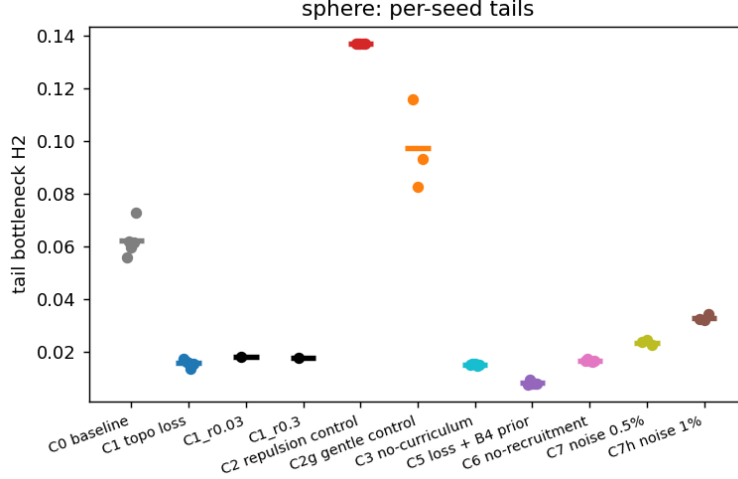


Figure 3: Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from *above*, so no tested repulsion strength reproduces any of the gain; the ρ ablation arms are indistinguishable from $\rho=0.1$.

five seeds (significant-count trajectories flat at the true $\beta_1=2$, Fig. 4), while the control collapses one loop. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not, and the reason is dimensional: a loop has minimal spatial support — it is characterized almost entirely by its birth/death values — so allocation can only over- or under-tessellate its neighbourhood (manufacturing phantoms or thinning the tube), while value pressure moves, merges, or tightens the loop directly. Voids, whose support is the whole shell, respond to both channels — which is why the composition (C5) wins everywhere.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error $9.5\times$, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Together with the allocation-channel results this is now a statement about the feature class: component structure is helped by any spreading pressure, not by topological information. The mechanism is that β_0 structure is carried entirely by the count and spatial separation of shells — a property *any* spreading pressure improves by default, and one the photometric term already infers from images. Topological guidance earns its keep where geometry and topology decouple: loops and voids.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 ($7.6\times$ vs C0, $\approx 2\times$ vs C1 alone, $\approx 4.6\times$ below the prior alone), torus .0133 — with the best sphere Chamfer as well; on the torus its Chamfer (.502) trails only the no-ramp arm (.495). Allocation and value-tuning are complementary mechanisms, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study’s floor, Appendix A), and they dominate both diagram metrics regardless of guidance. The loss adds no phantoms there (β_1 -significant count = 2 in every run) and its Chamfer is marginally

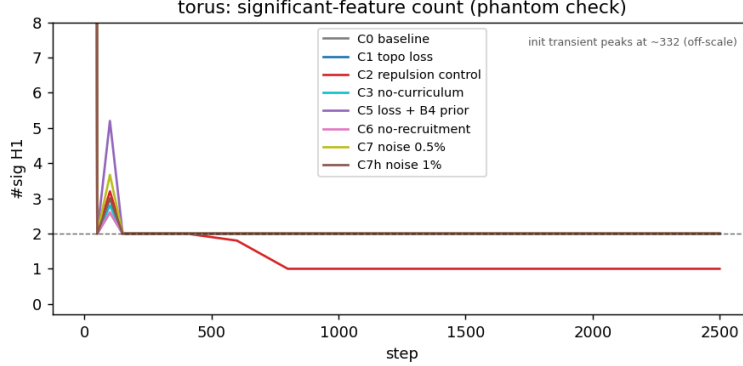


Figure 4: Torus phantom check: significant-H1 count over training (seed means; dashed line: true $\beta_1=2$). After a brief resampling transient near step 100, every loss arm sits exactly on two significant loops for the rest of training — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~ 800 and never recovers it. All arms share the off-scale random-init transient (annotated).

best; the case measures the *representation’s* ceiling, not the channel’s — both floors (measurement and representation) are quantified in §6.

Ablations. ρ -response flat over $\{0.03, 0.1, 0.3\}$ (.0181/.0172/.0177 on sphere); C3 (no curriculum) $\approx$ C1 on both decisive shapes *and* off the analytic family — on the genus-1 external mesh C3 lands at $.0171 \pm .0026$ vs C1’s $.0208 \pm .0008$, so the curriculum-free finding generalizes. ≈ 46 s per 2,500-step run (refresh 174 ms at $M=2048$ every 10 steps) — on these deliberately tiny scenes that exceeds the 33–41 s photometric baseline itself, but the cost scales with M ($O(M \log M)$ per refresh), not with scene or image complexity, so it amortizes on realistic scenes where rendering dominates; the loss adds no trainable parameters.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: $.0411 \pm .0038$, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer — though the loops still *exist* (#sig H1 = 2 in every seed; Fig. 2, right: the C6 curve rides the baseline). The run logs locate the mechanism precisely: at the loss’s own sampling density, one torus loop is significant at every refresh while the other sits chronically *below* the significance threshold — in all five seeds, at all 200 refreshes — so matching and diagonal pressure act only on the loop the live diagram already measures, and recruitment is the sole gradient path to the other. It is not a cold-start device; on this class it *is* the mechanism. On the sphere the void is always live and matched, recruitment never fires (0 of 200 refreshes, all seeds), and C6 is loss-identical to C1 by construction — its tail (.0166 vs C1’s .0156, 1.5σ) doubles as a measurement of pure run-to-run (CUDA) noise.

Sensor-noise stress (C7/C7h). With the plan’s cloud perturbed at every refresh ($\sigma = 0.5\%/1\%$ of the diagonal $\approx 0.5/1.0$ median sample spacings, §4), the loss degrades gradually and never harms: torus $.0210 \pm .0018$ / $.0259 \pm .0022$ ($2.0\times/1.6\times$ below baseline, vs C1’s $2.3\times$), sphere $.0235 \pm .0009$ / $.0328 \pm .0013$ ($2.7\times/1.9\times$, vs $4.0\times$) — at equal-or-better Chamfer, with the correct significant-feature count in every run (Fig. 2). The internals explain the grace: across all 2,400 noisy refreshes *not one* significant pair failed the Gabriel gate — the detachment fallback of §3 was never needed

Table 2: Generality wave: tail bottleneck to target (mean $\pm$ sd, 3 seeds); $\times$ = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521 $\pm$.0081	.0237$\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)	PASS (6.1 σ)
bob (H1)	.0426 $\pm$.0012	.0208$\pm$.0008 (2.1 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)	PASS (26.1 σ)
fandisk (H2)	.0538 $\pm$.0022	.0052$\pm$.0008 (10.4 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)	PASS (35.7 σ)
tom-yum pot (H2)	.0221 $\pm$.0003	.0057$\pm$.0002 (3.9 $\times$)	.0223 (β_2 1 $\rightarrow$ 0)	PASS (80.1 σ)

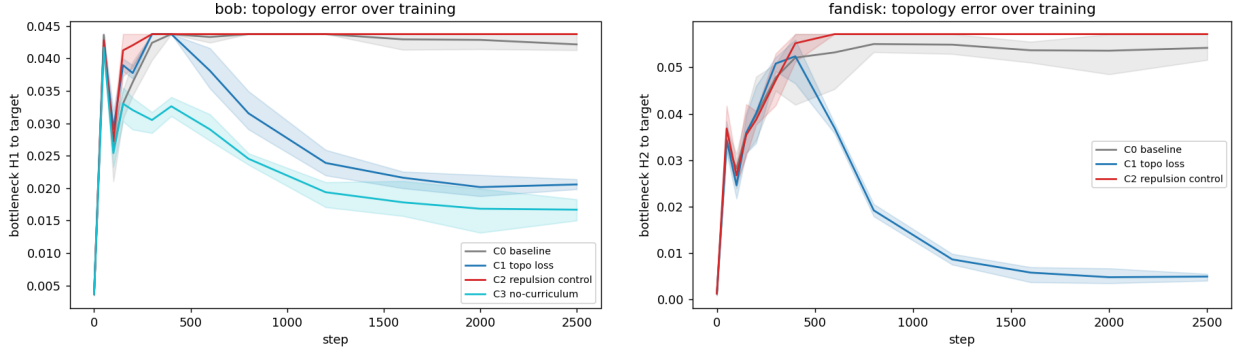


Figure 5: Generality trajectories (line = seed mean, band = seed min-max). *Left*: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. *Right*: fandisk (CAD) — the largest effect measured in this study (10.4 $\times$).

— and recruitment absorbed the noise: at $\sigma=1\%$ the torus’s second loop is sub-threshold at every refresh and recruitment reaches it every time (zero unreached targets throughout). Calibration is the noisiest part (λ_{peak} spreads $\approx 3\times$ across seeds), yet tails stay tight — consistent with the flat ρ response.

5.3 Generality: external genus-known meshes

Everything above is on one analytic scene family. We therefore repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, sampling density fixed by the pre-registered floor rule (the 2048 default everywhere except the tom-yum pot’s 8192, §4) — on four external meshes of known genus (§4): *spot* (genus 0, organic), *bob* (genus 1, organic), the *fandisk* CAD benchmark (genus 0), and the *tom-yum pot* (genus 3, artist-authored, certified by ingest). No per-shape tuning of any kind: the headroom budgets transfer from the feature class, bob runs the full unrestricted bundle, and the pot’s density is dictated by the rule, not chosen. All four **pass** the verdict rule at equal-or-better Chamfer (Table 2, Fig. 5), and the control’s failure modes replicate exactly: repulsion erases the void on all three H2 shapes (β_2 1 $\rightarrow$ 0, all seeds; on the tom-yum pot at 1.36 $\times$ worse Chamfer) and collapses a loop on the genus-1 shape (β_1 2 $\rightarrow$ 1, all seeds).

Four observations. *The H1 claim survives the family change*: bob halves loop error with zero phantom handles in every seed (26.1 σ), replicating torus. *Sharp-featured CAD benefits most*: fandisk’s 10.4 $\times$ exceeds every analytic-family effect. *Complex organic geometry exposes a count-level limit*: spot’s *baseline* manufactures a phantom void in two of three seeds (final #sig H2 = 2, 1, 2); the loss clears it in two seeds but retains one spurious bar in the third (1, 2, 1) — value repair is reliable, count repair on sub-budget organic detail (legs, horns at $N=1200$) is an improvement, not

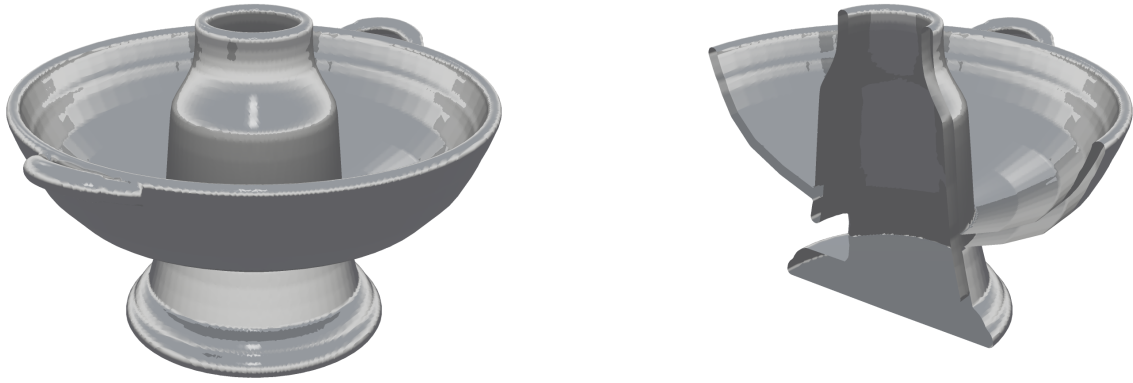


Figure 6: The *tom-yum pot*: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, $\beta=(1, 6, 1)$ exact (§4), rendered with an aluminium material. *Right*: cutaway. The narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber — the on-axis H2 void that the floor rule selects as the discriminating observable at $M=8192$; every H1 loop (handle bores, flue) sits below the measurement floor at all working densities.

a guarantee. Spot’s C1 tail is also pinned across seeds (sd 0 at .0237), suggesting a geometric floor of the same kind as the double torus, far milder in degree. *And the pipeline reaches raw artist content*: the tom-yum pot passes the same rule with its observable and sampling density chosen by the pre-registered floor rule rather than by hand — the dress rehearsal for scan data, whose ground truth must likewise be certified, not assumed (§4).

From artist soup to verdict (the tom-yum pot). The certified artist mesh (Fig. 6) extends the wave in two directions at once: the input class (a raw Blender export, certified rather than constructed, §4) and the observable (an H2 void that only clears the measurement floor at $M=8192$). The loss cuts the void’s value error $3.9\times$ at Chamfer parity ($0.99\times$ baseline); the 80.1σ Welch statistic reflects an unusually tight seed spread at $n=3$ (C1 sd .0002), so the honest summary is *non-overlapping seed ranges*, with the effect size inside the study’s $2.1\text{--}10.4\times$ span. The control *destroys* the void ($\beta_2 1\rightarrow 0$ in all three seeds) at $1.36\times$ worse Chamfer; its bottleneck pins at the unmatched-target bound (the target bar’s half-life, .0223), so the count — not the meaningless value gap (0.8σ) — is the verdict. Two caveats transfer verbatim from the rest of the study: the baseline already reads the correct count (a value-accuracy win, like bob), and ground truth here is certified, not constructed.

6 Mechanism and discussion

Sparse tugs vs. budget allocation. Each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices, so the loss delivers *precise but sparse* pulls, iterated by re-pairing — the toy probes show surgery “walking” around a defect a few points at a time; the prior, by contrast, biases every respawn decision diffusely across a feature’s whole support. The resampling prior is the opposite: broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which the prior could only over- or under-tessellate, respond to precise value pressure (H1 win); wide 2-D shells benefit from both (H2: prior helps, loss helps more, composition best); 0-D component structure needs neither — any spreading pressure suffices (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface; the void’s death time pins at that equilibrium (identical across seeds) and thin structures collapse (β_2 loss on the cube, β_1 loss on the torus). Fig. 7 makes this visible: once the repulsed cloud reaches its equilibrium spacing, no refinement can tighten the void closure — the death coordinate is pinned, then dragged, while near-diagonal phantom bars accumulate. The loss has no such length-scale constraint: it pulls the death coordinate itself and the geometry follows, rather than the other way around. This sharpens the control argument: spending the same gradient budget through the same channel *without* topological information is not neutral — it is harmful — so C1’s wins cannot be an artifact of extra regularization.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its sampling density M : the torus void and the double torus’s two tube loops do not, at $M=2048$. We treat this as a principle rather than a nuisance: the target bundle is built (and audited) at the same M , features below the floor are excluded from the objective, and the double-torus stretch case is evaluated against its two measurable loops — which is also all the reconstruction itself can express at feasible budgets. Raising M buys resolution at linear-ish persistence cost; the trade is explicit and logged at bundle-build time.

The floor is quantifiable. A bar is significant iff its lifetime exceeds $6r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour spacing, the implementation’s self-calibrating threshold), and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over $M=2048\rightarrow 20000$; ratio 3.16 vs $\sqrt{9.77}=3.13$). This puts numbers on both stories in this paper. The double torus’s tube loops live at lifetime .045–.046 on the clean surface: at $M=2048$ that is $0.83\text{--}0.85\times$ the floor — invisible by a 15% margin, clearing it only from $M\approx 3\times 10^3$ — while at the evaluation density ($M=20000$, floor .0171) they clear it $3.1\times$; the binding constraint there is the *representation* (the soup never expresses the tubes at any feasible N), so raising M alone cannot close the gap. The torus quantifies the interaction with reconstruction noise: its second loop clears the clean-surface floor by only $1.33\times$, a margin the live cloud’s noise consumes — which is exactly why that loop is chronically sub-threshold during training and recruitment becomes its only gradient path (the C6 ablation). Appendix B tabulates the counts across M ; at $M=4096$ all four double-torus loops clear the floor. The tom-yum pot ran the rule *prospectively*: on the certified shell no H1 loop clears the floor at any working M , while the flue void first clears it at $M=8192$ ($1.20\text{--}1.23\times$ across sampling seeds) — so the rule, not judgment, picked both the observable class and the bundle density before any training run (§4), and the in-loop result (Table 2) validated the choice. Adaptive or multi-scale density — raising M where features sit near the floor, or hierarchical floors — is the natural extension; the trade is explicit, since persistence cost grows $O(M\log M)$.

Recruitment is landscape-guided in practice — and it is the mechanism for loops. Its location blindness — pull the nearest bar anywhere — did not misfire on our geometry because unproductive tears self-extinguish (their bar’s death saturates at the local detour scale) while productive cuts run away toward the target. The caveat stands for target features larger than any achievable detour; the probe and its mechanism are reported so the term is used with eyes open. The C6 ablation sharpens its role beyond cold-start repair: a loop that never clears the live significance floor at M receives metric pressure *only* through recruitment, at every refresh — matching alone reduces the torus to baseline. Recruitment is how the loss reaches below its own significance floor on the live side.

Prescription. Correct topology in the *loss*; allocate budget with a *wide* prior; combine for the best of both. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7 Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU. The scene family is analytic plus four external genus-known meshes (§5.3; one of them a raw artist mesh ingested and certified — the intended path for scans) — still short of a public shape-set benchmark — and a real (open-surface, clinical) demonstration remains future work; open surfaces add a genuine research obstacle — scan boundaries contaminate H1, and no clean ground-truth diagram exists. Concrete paths exist: a boundary-aware persistence definition (relative homology), filtering boundary-born bars before matching, or template-aligned initialization that suppresses boundary noise; we judge bar-filtering the pragmatic near-term step, with a synthetic open-surface stress test (a ray-cast solid plus point noise) as the natural first experiment.

Density floor. The loss cannot see features below the significance threshold at its sampling density (torus void; two of four double-torus loops at $M=2048$). The bundle audit makes this

visible, not invisible — but it is a real ceiling on what the objective can enforce.

Fixed target diagram. The target bundle presumes known ground-truth topology (true in reconstruction-from-scans settings where a prior scan or template exists; not in blind capture). Three template-free designs are natural but untested here: minimizing total persistence (risks collapsing legitimate structure), enforcing a target *genus* via a penalty on the significant-H1 count (the most robust of the three, and the one we would try first for blind capture), or statistical priors over a shape class’s diagrams.

Control strength. The repulsion control is aggressive (it destroys topology at the calibrated magnitude); a gentler variant at half the radius is still $1.6\times$ worse than baseline (sphere: .0972 vs .0623), so the verdicts do not hinge on control tuning — but control *design* for loss-channel studies deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup’s CUDA training is not bit-reproducible: divergence enters at resampling, where borderline prune/split decisions flip under floating-point drift, so even baseline re-runs differ in detail. Quantitatively, tail coefficients of variation run 7–10% for baseline and loss arms alike, and a loss-identical re-run pair (the sphere C6 arm, which is C1 by construction there) landed 6% apart — the scale of pure run-to-run noise. All statements are therefore seed-averaged with Welch statistics.

Recruitment guarantees. Recruitment’s self-correction premise — unproductive edits saturate at the local detour scale — is validated empirically (toys, training, stress test), not proven; pathological geometries (very thin filaments, tightly constrained manifolds) could defeat it. A failure-case study and a hardness analysis of recruitment-based matching are future work.

8 Conclusion

This paper moved topology out of the resampler’s allocation channel and into the training objective of a differentiable triangle-soup pipeline. Four things carry the contribution: a matched persistence loss whose pair-frozen backward is *certified* per pairing by a Gabriel-exactness gate; a recruitment term that restores the gradient optimal matching provably lacks for missing features — and which an ablation shows is the load-bearing mechanism for the loop class; a channel-controlled evaluation in which every gain must beat a norm-matched non-topological control at Chamfer parity; and a quantified account of the method’s floors (the $6r_{\text{med}}(M)$ measurement floor and the budget- N representation ceiling). The verdicts replicate on external genus-known meshes — including a raw artist mesh ingested and certified by measurement — without per-shape tuning, and degrade gracefully under sensor noise. The prescription stands: correct topology in the *loss*, allocate budget with a *wide* prior, and combine them — the two channels are complementary, and the composition is best everywhere tested. Future work: open surfaces and real scans (boundary-aware persistence), adaptive or multi-scale sampling density, template-free targets (genus-level penalties), and curvature-weighted spatial masking of the loss.

A The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer

equal by construction (40k samples, metrics normalized by the ground-truth bounding diagonal; Table 3). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff₉₅ actually *prefers* the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

Table 3: Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Chamfer% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

(2) The prior arms, and why voids are “width-primary”. The study biases DiffSoup’s own resampler with a precomputed importance field built from the target’s significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature’s own death scale:

$$\phi_d(q) = \sum_{f: \dim f=d} \sum_{c \in C_f} w_f e^{-\|q-c\|^2/(2\sigma_f^2)}, \quad \sigma_f = s \operatorname{clip}(\sqrt{\text{death}_f}, 3r_{\text{med}}, \tfrac{1}{2}\text{diag}), \quad w_f = p_f/s^2, \quad (5)$$

evaluated on a dense target-surface sample, normalized per dimension to $[0, 1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample’s value. The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \rightarrow s\sigma$, $w \rightarrow w/s^2$, so $w\sigma^2$ is s -invariant): $s=1$ is the concentrated prior (B2), $s=3$ the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of $\text{vis} + \lambda Q_{0.9}(\text{vis}) \cdot \text{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. (Table 4; $N=1200$, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ , sphere 2.2σ , cube tie); one-shot placement washes out entirely ($B1 \approx B0$, .0546 vs .0535 on the sphere). Hence the verdict quoted in §1: the prior’s topological value is carried largely by its spatial *width*.

Table 4: Allocation study, void class (H2), $N=1200$, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

shape	B0	B2 concentrated	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

(3) No prior shape repairs loops. On the torus ($N=700$, five paired seeds, true $\beta_1=2$) every arm fails the topology-specificity test: the best condition is the *non-topological* wide control B5

(.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) *and* manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of this paper, and the class the loss channel wins (§5.2, Table 1).

B Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table 5 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6r_{\text{med}}(M)$ of §6. It confirms the three density claims made in the paper: the sphere’s void clears the floor at every tested density; the torus needs $M=2048$ before its *second* loop is measurable, and its own void becomes measurable only at $M=4096$ (hence the H1 restriction at $M=2048$); and the double torus’s tube loops cross the floor between $M=2048$ and $M=4096$ — consistent with the $M \gtrsim 3 \times 10^3$ estimate of §6 — so all four loops are measurable at $M=4096$.

Table 5: Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M ; floor $= 6r_{\text{med}}(M)$, seed 0.

	$M=512$	$M=1024$	$M=2048$	$M=4096$
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

Diagram trajectories: the pinning mechanism, visualized. Fig. 7 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles $\approx .06$ below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within $\approx .016$ of target — matching its measured tail. The control arm is qualitatively different: the void’s death is dragged progressively down ($\approx .23 \rightarrow .15 \rightarrow .13$) while a carpet of phantom mid-persistence bars accumulates near the diagonal — the repulsion-equilibrium signature discussed in §6.

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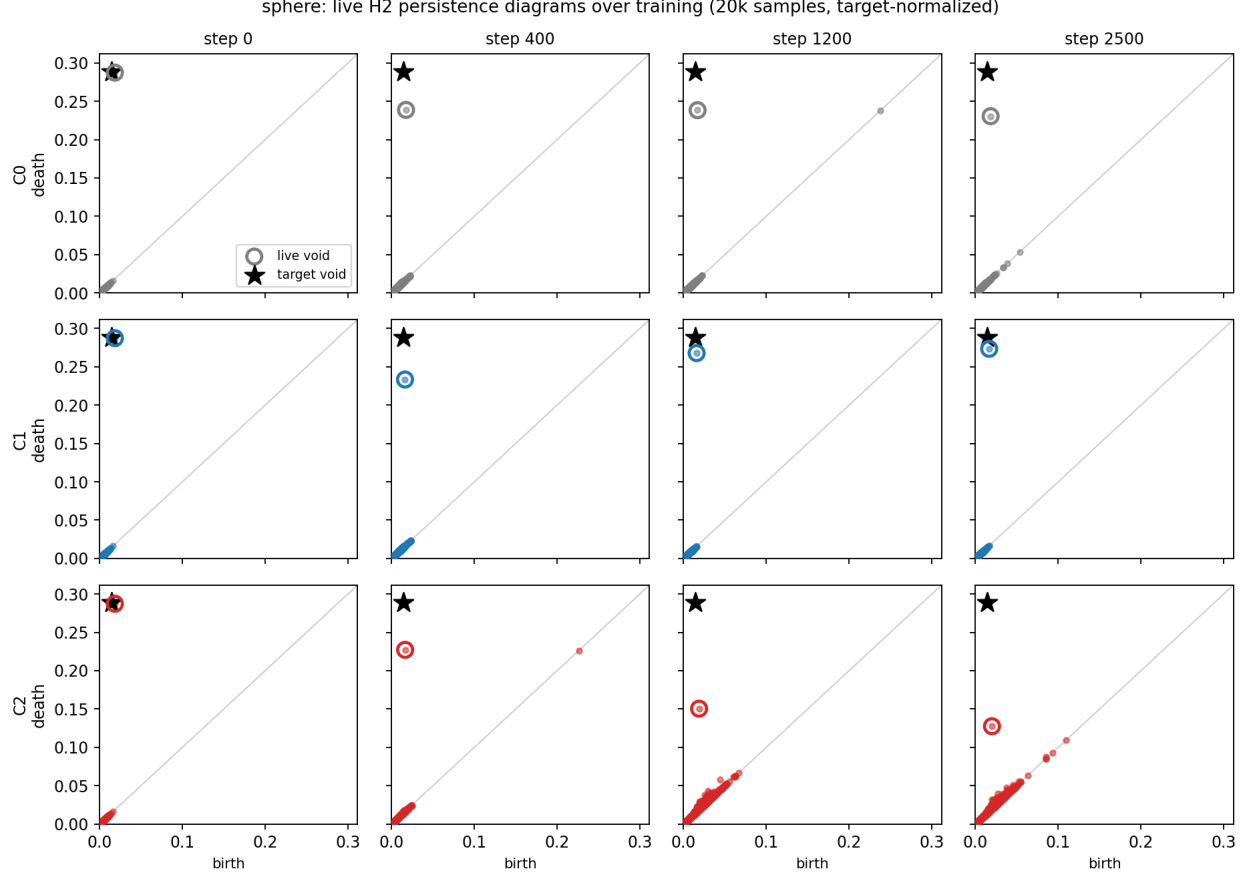


Figure 7: Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). *Top*, baseline: the void drifts off target. *Middle*, loss: the void returns to the target once the ramp activates. *Bottom*, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

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